

Conversion-Prediction Baseline — Deployment Readiness Assessment

Model Card v1.0 | UNC-768

ML Engineer, Uncle May's

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1 Executive Summary

Verdict: NOT READY FOR PRODUCTION. The conversion-prediction prototype in `01_first_look_baseline.qmd` is a well-structured research notebook, not a deployable system. There is no serialized model artifact, no serving API, no monitoring, and no retraining loop. The training dataset (305 rows, 10 real positives before anon-capping) is too small to produce estimates with confidence intervals narrower than ± 0.05 AUC. The correct near-term decision is to **continue collecting data and investing in attribution coverage** before deploying to any real decision (ad bidding, personalization, or segmentation).

This document:

1. Summarizes the prototype as built
2. Scores it against a deployment readiness checklist
3. Ranks the gaps that must be closed before production use
4. Maps the data dependencies the model requires

2 1. Model Summary

2.1 1.1 Business Question

Given a checkout attempt (a Stripe Checkout Session or PaymentIntent), what is the probability that this customer completes payment?

Intended applications: abandon-recovery prioritization, ad-bid adjustment, customer segmentation. **None of these applications are live yet.**

2.2 1.2 Architecture

The prototype evaluates **six (model × resampling) combinations** via 5-fold stratified cross-validation:

Model	Engine	Resampling	Imbalance handling
Logistic Regression (L1)	glmnet (R)	Balanced	<code>case_weights</code> via workflow
Random Forest	ranger (R)	Balanced	<code>class.weights = c(0=1, 1=pos_weight)</code>
XGBoost	xgboost (R)	Balanced	<code>scale_pos_weight = neg/pos</code>
Logistic Regression (L1)	glmnet (R)	SMOTE	synthetic minority oversampling
Random Forest	ranger (R)	SMOTE	synthetic minority oversampling
XGBoost	xgboost (R)	SMOTE	synthetic minority oversampling

Hyperparameters are fixed (not tuned): LR penalty = 0.01, RF trees = 500, XGB trees = 500 / lr = 0.05 / depth = 4. No hyperparameter search has been run.

2.3 1.3 Training Data

Metric	Value
Source parquet	<code>conversion_v2_20260504T033429Z.parquet</code>
Total rows (raw spine)	305
Columns	93
Positive class (converted)	10
Negative class (abandoned)	295
Raw base rate	3.3%

Metric	Value
Rows with non-null email_hash	25

Why the notebook shows ~34% positive rate but the raw parquet is ~3%: `clean_attempts()` in `_feature_prep.R` caps anonymous (null-email) attempts at 3 per fingerprint bucket, collapsing the bot/probe-dominated negative class. This is intentional — the model should learn “what predicts conversion among real-looking attempts,” not “is this anonymous traffic?” The cleaning step is correctly designed but dramatically changes the effective dataset size.

2.4 1.4 Feature Space

Table 3: Feature sources — 75 total features (before recipe drops)

source	n_features
Stripe / Core	13
GA4	10
Census ACS	7
Google Ads	7
Meta Ads	7
Attempt Features	6
Mailchimp	6
Checkout Store	5
Stripe Customer	5
Apollo	4
Other	2
Resend	2
Campaign Map	1

Recipe transformations (from `_feature_prep.R`):

- Log-transforms `amount_cents` and `seconds_since_prior_attempt`
- Extracts day-of-week, hour, and is-weekend from `created_at`
- Lumps rare categoricals (<2%) into “Other”
- Drops zero-variance and near-zero-variance columns
- Median-imputes numeric predictors (for `glmnet` compatibility)
- SMOTE applied post-recipe when `use_smote = TRUE`

2.5 1.5 Held-out Performance

Note: The baseline notebook does not produce a held-out test-set score. All metrics reported below are 5-fold cross-validation estimates from `01_first_look_baseline.qmd`. With only 10–30 positives depending on the anon-capping threshold, each fold contains 2–6 positive examples. AUC estimates at this sample size have standard errors large enough that model comparisons are unreliable. The PR-AUC numbers should be treated as rough feasibility signals, not deployment thresholds.

Metric	Interpretation	Reliability at N 30 positives
ROC-AUC	Rank-ordering quality	Low (± 0.10 SE expected)

Metric	Interpretation	Reliability at N 30 positives
PR-AUC	Precision-recall tradeoff	Very low (± 0.10 – 0.15 SE)
Brier score	Calibration of probabilities	Low

Majority-class PR-AUC floor (= base rate after anon-capping): approximately **0.34**. Any model must beat this to add value.

The Decision Scientist’s notebook summary (section 8) states directly: “*N=X with Y positives is too thin for a deployable model. Treat these PR-AUC numbers as ± 0.05 noise.*”

3 2. Deployment Readiness Checklist

Table 5: Deployment readiness checklist — conversion prediction v1

Category	Item	Status	Notes
Artifact	Serialized model file (.pkl / .joblib / .rds)	MISSING	No artifact found in <code>ml/models/</code> . Notebook trains but never saves.
Artifact	Reproducible training script (separate from notebook)	MISSING	Training logic is embedded in the Quarto notebook. No <code>train.py</code> or equivalent.
Artifact	Versioned artifact registry (S3 / GCS / MLflow)	MISSING	No registry. Local disk only.
Serving	Prediction API endpoint (REST / Trigger.dev task)	MISSING	No API. No Trigger.dev task. No R plumber service.
Serving	Batch prediction pipeline (offline scores in BQ / parquet)	MISSING	Not built.
Serving	Input validation (schema enforcement at inference time)	MISSING	No serving layer exists to validate against.
Monitoring	Score distribution drift alert	MISSING	No monitoring infrastructure exists.
Monitoring	Feature drift alert (population stability index or similar)	MISSING	Not built.
Monitoring	Ground-truth feedback loop (labels landing back to evaluate live predictions)	MISSING	Not built. Labels are only visible after Stripe resolves the attempt.
Retraining	Retraining trigger (schedule or performance threshold)	MISSING	No trigger. Retraining is manual.

Cate- gory	Item	Sta- tus	Notes
Re- train- ing	Automated pipeline to rebuild <code>conversion_v2</code> parquet and re-fit	PAR- TIAL	<code>run_pipeline.py</code> and <code>ml/ingest/*.py</code> exist; no orchestrated retrain schedule.
Data	Stable, versioned feature parquet produced reliably by Data Engineer	PAR- TIAL	Ingest scripts exist and ran on 2026-05-02. No reliability SLA or freshness alert.
Data	GA4 attribution coverage >10% (needed for GA4 features to add signal)	PAR- TIAL	Attribution fixed 2026-05-03 (UTM/gclid/fbclid now flow end-to-end). Historical coverage <1%. Future coverage TBD.
Data	Minimum training positives (100 real converters recommended)	MISS- ING	10 positives in raw parquet (2026-05-02). ~30 post anon-capping. Well below threshold.
Gov- er- nance	Model card reviewed and approved by Decision Scientist + CEO	MISS- ING	This document is the first model card. Pending review (UNC-768).
Gov- er- nance	Fairness check (disparate impact on protected characteristics)	MISS- ING	Census ACS demographic features are in the feature set. No fairness audit done.
Gov- er- nance	Retraining cadence and on-call owner defined	MISS- ING	Not defined.

Score: 0 / 16 items fully done, 3 partial.

4 3. Gaps and Recommendations (Ranked)

Items are ranked by blocking severity: **P0** = must fix before any production signal is used; **P1** = must fix before soft-launch to a single use case; **P2** = should fix within 30 days of soft-launch.

4.1 P0 — Blockers (do not deploy until resolved)

P0-A: Training N is too small. 10 raw positives (30 post anon-capping) yields cross-validation estimates with ± 0.05 – 0.15 AUC standard errors. Confidence intervals span most of the meaningful range. A model trained on this data will assign scores with very wide uncertainty, making it unsuitable for bidding or personalization.

Recommended action: Wait until the dataset has 100 real converters (not SMOTE-augmented). At the current trajectory, this requires monitoring weekly ingest from Stripe and re-evaluating after each weekly delivery cycle. Custom- cart launch will accelerate this.

P0-B: No serialized model artifact. Even if metrics were acceptable, there is nothing to serve. The notebook trains models in memory and discards them at session end.

Recommended action (when N is sufficient): Add a `save_model()` step at the end of the training notebook or in a standalone `ml/models/train.py` script. For R, use `saveRDS(fit_rf, "ml/models/conv_pred_v1_rf.rds")`. For Python portability, serialize with `joblib` after porting the winner to scikit-learn.

P0-C: No fairness audit. The feature set includes Census ACS demographic proxies (income, age, household size, racial composition by ZIP). Using these to rank customers for recovery or bidding requires an explicit disparate-impact audit. This is both a legal and ethical blocker.

Recommended action: Before any deployment, run a fairness analysis (AUC by ZIP-income quartile, by racial composition quartile). Document the analysis in a companion Quarto notebook. Escalate to CEO if disparate impact is found.

4.2 P1 — Required for soft-launch

P1-A: No serving layer. There is no API, no Trigger.dev task, and no offline batch job that generates scores. Predictions cannot reach any downstream system.

Recommended action: Once an artifact exists, build the simplest viable serving path. For Uncle May’s at current scale, this is likely a Trigger.dev TypeScript task that:

1. Reads the latest `conversion_v2` parquet or calls a Python subprocess
2. Calls `predict_proba()` on the serialized model
3. Returns a JSON `{ attempt_id, score, threshold_flag }`

Alternatively: an R `plumber` API containerized in Docker on Cloud Run. Discuss the trade-off with the Decision Scientist before choosing.

P1-B: No ground-truth feedback loop. Predictions are only useful if we can later compare them to actual outcomes. Without this loop, model monitoring is impossible.

Recommended action: Log every prediction (`attempt_id`, `score`, `timestamp`, `model_version`) to a BigQuery table. The Stripe webhook already fires on payment success/failure — join that signal back to the prediction log weekly.

P1-C: GA4 attribution coverage below threshold. GA4 session features (pageviews, scroll depth, form starts, traffic source) are the most valuable behavioral signals the model could use. As of 2026-05-02, coverage was <1% (1 of 305 rows had a GA4 match). Attribution was fixed on 2026-05-03, but historic data cannot be backfilled. GA4 features will only become reliable after several weeks of post-fix data accumulates.

Recommended action: Track GA4 join rate weekly as a data quality metric. Do not use GA4 features as load-bearing in a production model until join rate exceeds 30% (90 matched rows at 300-row scale, or 300 rows at 1,000-row scale).

4.3 P2 — Within 30 days of soft-launch

P2-A: Hyperparameter tuning. Fixed hyperparameters are a research shortcut. Before committing to a model architecture, run `tune_grid()` or `tune_bayes()` on the top-2 candidates.

P2-B: Monitoring dashboard. Weekly: plot score distribution of new predictions vs. the training distribution. Flag if KL-divergence exceeds a threshold. This can be a short Quarto document auto-rendered by a Trigger.dev cron.

P2-C: Retraining cadence. Define the trigger: “retrain weekly if new data adds >20 positives since last retrain” OR “retrain on the 1st Sunday of each month.” Document the owner (ML Engineer) and the on-call escalation path.

P2-D: Separate train/test split. Current evaluation is 5-fold CV only — no holdout. Once N = 100 positives, do a temporal holdout (most recent 20% of data by `created_at`) to get an unbiased estimate of live performance.

5 4. Data Dependency Map

The `conversion_v2` feature pipeline requires the following raw parquet files, all produced by `ml/ingest/*.py`:

Table 6: Data dependencies for `conversion_v2` feature pipeline

Source	Ingest_Script	Join_Key	Features_Added
Stripe Checkout Sessions	<code>in-gest/stripe.py</code>	<code>checkout_session_id</code>	Core attempt spine (label, amount, timestamps, UTMs)
Stripe Payment Intents	<code>in-gest/stripe.py</code>	<code>payment_intent_id</code>	Core attempt spine (label, amount, timestamps, UTMs, fbc/fbp/gclid)
Stripe Customers	<code>in-gest/stripe.py</code>	<code>customer_id</code>	Billing geo, delinquency flag
Stripe Charges	<code>in-gest/stripe.py</code>	<code>payment_intent_id</code>	Charge outcome columns (leakage-flagged — currently dropped in recipe)
Stripe Invoices / Subscriptions	<code>in-gest/stripe.py</code>	—	Not used in v2 model (subscription flow excluded from label)
GA4 Session Summary	<code>ingest/big-query_ga4.py</code>	<code>ga_client_id</code> / <code>payment_intent_id</code>	Device, browser, traffic source, pageviews, scrolls, form_starts
Mailchimp Members	<code>in-gest/mailchimp.py</code>	<code>email_hash</code>	Subscription status, open/click rates, tag count
Resend Emails	<code>ingest/re-send.py</code>	<code>email_hash</code>	Send count, email type count
Apollo Contacts	<code>in-gest/apollo.py</code>	<code>email_hash</code>	Professional title, seniority, industry
Census ACS (ZIP)	<code>ingest/census.py</code>	<code>shipping_zip</code>	Median income, population, age, household size, racial composition
Meta Campaign Insights	<code>in-gest/meta_ads.py</code>	<code>campaign_slug_canonical</code>	Lifetime impressions, clicks, spend, CTR, CPC per campaign
Google Ads Campaign Insights	<code>in-gest/google_ads.py</code>	<code>campaign_slug_canonical</code>	Lifetime impressions, clicks, spend, conversions, CTR, CPC
Checkout Store (local)	<code>ingest/checkout_store.py</code>	<code>payment_intent_id</code>	Cart line count, total, fulfillment mode, recovery email flags
Microsoft Clarity	<code>ingest/clarity.py</code>	—	URL-level dwell/scroll/rage-click (NOT joined in v2 — noted as skipped)

5.1 Reliability assessment

Table 7: Ingest reliability status (2026-05-03)

Source	Reliability
Stripe Checkout Sessions	Live
Stripe Payment Intents	Live
Stripe Customers	Live
Stripe Charges	Live
Stripe Invoices / Subscriptions	Live
GA4 Session Summary	<1% join rate (pre-2026-05-03); improving post-fix
Mailchimp Members	Live

Source	Reliability
Resend Emails	Live
Apollo Contacts	Live
Census ACS (ZIP)	Annual; refresh once/year
Meta Campaign Insights	Live
Google Ads Campaign Insights	Live
Checkout Store (local)	Partial — only post-2026-04-24 data
Microsoft Clarity	Available but not used

Key observations:

1. **GA4 join rate is the critical gap.** All GA4 behavioral features are functionally null for the entire historical dataset. Post-fix data will accumulate from 2026-05-03 forward.
2. **Checkout Store is partial.** Only attempts from 2026-04-24 onward have cart-level features (line count, cart total, recovery flags). These may be among the most predictive features when sufficient history accumulates.
3. **No freshness SLA exists.** Ingest scripts are run manually. A broken ingest (e.g. Stripe rate limit, GA4 token expiry like the 48h gap on 2026-04-29) can silently produce stale features. Recommend a daily freshness check: if any raw parquet is >36h old, alert the Data Engineer.
4. **Census ACS** is low-risk because it refreshes annually. Confirm the ACS vintage year and document it.

6 5. Recommended Next Steps (Prioritized)

Priority	Owner	Action	Precondition
P0-A	Data Engineer + DE Scientist	Monitor weekly: track cumulative positives in Stripe; surface when N_pos > 50, 100	—
P0-C	Decision Scientist	Fairness audit on Census ACS demographic features	Can run now on current data
P0-B	ML Engineer	Add <code>saveRDS()</code> / <code>joblib.dump()</code> to training notebook	N_pos 100
P1-B	ML Engineer	Create BQ prediction log table; log every future score	Serving layer
P1-A	ML Engineer	Build minimal serving layer (Trigger.dev task or R plumber)	Model artifact
P1-C	Data Engineer	Track and report GA4 join rate weekly	Attribution fix live

Priority	Owner	Action	Precondition
P2-A	Decision Scientist	Hyperparameter tuning (tune_grid / tune_bayes)	N_pos 100
P2-B	ML Engineer	Weekly score-distribution monitoring Quarto	Prediction log
P2-D	Decision Scientist	Temporal train/test split evaluation	N_pos 100

7 6. Reviewer Sign-Off

This document is submitted for review under **UNC-768**. Required reviewers:

- ☐ **Decision Scientist** — validate model summary accuracy and gap prioritization
- ☐ **CEO (Anthony Ivy)** — acknowledge readiness verdict before any production deployment decision

No model deployment actions will be taken until both reviewers sign off.

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